

Gallery + Sharing Tutorial

Publish a design. Remix someone else's. Save a fastener once and reuse it everywhere. Pick a license you actually want to live with.

The Gallery and Component Library are ForgeSlicer's two community spaces. The Gallery is for whole designs (a phone stand, a wall hook). The Library is for reusable parts (an M3 socket-cap screw, a hinge). They share the same publish flow.

1. Anatomy of a shared item

Every gallery card is a packaged bundle stored on ForgeSlicer's server. The bundle contains:

Field	Required?	Notes
Name	Yes	Shown on every card; max 80 chars; supports emoji.
Thumbnail	Auto	Captured from the viewport at publish time. You can re-capture from the gallery card later if you don't like the angle.
Description	No	Markdown supported. Use it to explain print orientation, intended infill, and any assembly steps.
STL	Auto	Baked geometry — what most users will print.
Editable project (.forge.json)	Auto	The original primitives & modifiers, so remixers can edit the source, not just the mesh.
License	Yes	Pick at publish time. See section 4 for the full list.
Material hint	No	Suggested material (PLA, PETG, ABS, TPU, PA-CF). Filterable on the gallery grid.
Category	No	Tools, Toys, Tabletop, Lighting, Storage, Functional, Decorative, Misc. Filterable.
Tags	No	Free-form. Each tag becomes a clickable filter pill.
Bounding box	Auto	Used by 'Resize to my bed' to scale on remix import.
Private	Toggle	On = visible only to you. Useful for works-in-progress.

2. Publishing a design

- 1 Click **Share Design** in the top toolbar. A modal appears with a fresh thumbnail of your current view.
- 2 Fill in the Name (required), Description (optional), Category, Tags, Material hint, License.
- 3 Pick the Privacy mode: **Public** (default, shows up on the global feed) or **Private** (hidden, only you see it on the 'Mine' filter).
- 4 (Optional) Re-capture the thumbnail by rotating the viewport to a hero angle and clicking the camera icon in the modal.
- 5 Hit **Publish**. The bundle (thumbnail + STL + .forge.json) is uploaded; a success toast links to the new card.

Editing after publish: click any of your own cards → Edit. You can change description, tags, license, and privacy without affecting the geometry. Replace the geometry (re-share new version) by deleting the old card & publishing again.

3. The Component Library

Components are *reusable parts*: a screw, a hinge, a snap-fit clip. You build them once, save them, then drop them into any future project.

Save a component

- 1 Select the part(s) you want to save (or leave selection empty to save the whole scene).
- 2 Toolbar → **Component** button (next to Share Design). A focused modal appears — same fields as a design plus an *Origin offset* field.
- 3 **Origin offset** shifts the component's local origin so it lands flush to the bed every time you drop it. Defaults to the bounding box bottom — leave it.
- 4 Pick whether it's a **positive** assembly (e.g. a knob you'll add to a project) or a **negative** template (e.g. a screw clearance hole you'll subtract).
- 5 Publish. The card appears in **Gallery** → **Components tab**.

Add to project

Browse Gallery → Components tab. Each card has an **Add to project** button. The component drops in at the world origin, modifier set per its publish settings — instant assembly. Use the gizmo to place it.

Verified ✓ badge

Some components carry a green **Verified ✓** badge. These are community-vetted parts — ISO-correct fasteners, tested fits, accurate tolerances. You can request verification for your own component via **Profile** → **Components** → **Request verification**; a maintainer reviews within a week.

4. Licensing — what you actually grant

Picking a license is the most important decision when you publish. It tells everyone else what they can do with your design. ForgeSlicer offers nine options; here's a one-line summary of each.

License	Allow commercial?	Allow derivative?	Share-alike?
CC-BY 4.0 (<i>default</i>)	Yes	Yes	No
CC0 (<i>public domain</i>)	Yes	Yes	No
MIT	Yes	Yes	No
Apache 2.0	Yes	Yes	No (patent grant)
GPL v3	Yes	Yes	Yes (copyleft)
LGPL v3	Yes	Yes	Partial
AGPL v3	Yes	Yes	Yes + network
CC-BY-SA 4.0	Yes	Yes	Yes
CC-BY-NC 4.0	No	Yes	No
CC-BY-ND 4.0	Yes	No	—
ForgeSlicer Standard	Yes (digital only)	Yes	No

If you don't care: stick with the default **CC-BY 4.0**. Anyone can use your design for any purpose as long as they credit you. Anything more restrictive (NC, ND) reduces remix uptake significantly — community data shows ~70% fewer remixes for NC-licensed designs.

Every gallery card shows a license chip you can click to read the full plain-English summary. The full legal text is available at creativecommons.org / opensource.org.

5. Remixing — building on others' work

Every public design has a **Remix** button on its gallery card. Click it to load the original editable project into your workspace — primitives, modifiers, groups, and all.

Remix attribution

When you publish a remix, ForgeSlicer auto-fills a '*Remixed from ...*' line in your description, linking to the original card. You can edit or delete it — but most licenses (especially CC-BY) **require** you to keep the attribution. Don't strip it without permission.

Resize to my bed

If the original was designed for a 300×300 printer and yours is 220×220, click **Resize to my bed** on the card before remixing. The component is uniformly scaled so its bounding box fits inside your active printer profile with a 5 mm margin.

6. Finding things in the gallery

The Gallery has four filter controls plus full-text search:

Filter	Filters by
Category	Tools / Toys / Tabletop / Lighting / Storage / Functional / Decorative / Misc
Material	PLA / PETG / ABS / TPU / PA-CF (publisher's suggested hint)
License	Permissive (CC0, CC-BY, MIT, Apache, ForgeSlicer Standard) or Copyleft (GPL/LGPL/AGPL/SA) or Restricted (NC/ND)
Public / Mine	Only available when signed in. Mine includes your private items — they show a small lock badge.
Search box	Searches name, description, and tag text. Min. 2 chars.

Author profile pages

Click any 'by ...' author name on a card to visit their public profile (/u/<userId>). You'll see everything they've chosen to share — avatar, location, contact link — plus their full grid of public designs and components.

7. Troubleshooting

Symptom	Likely cause	Fix
Publish button greyed out	You're signed out, or scene is empty.	Sign in (top-right). Add at least one part before publishing.
Thumbnail is dark / wrong angle	Captured before you framed the part.	Move/rotate the viewport, then re-capture from the publish modal's camera icon.
Card shows 'private' lock but I want it public	Privacy toggle was on at publish time.	Edit the card → flip the toggle — no need to re-upload.
Remix loads as a flat mesh, can't edit primitives	Original was published without the editable <code>.forge.json</code> (rare — legacy items).	You'll have to slice or rebuild from the STL; ask the author to republish with the source.
'Resize to my bed' button missing	No active printer profile selected.	Right panel → Printer → pick a profile.

8. Quick reference

Toolbar buttons: *Share Design* (designs) and *Component* (reusable parts).

Gallery URL: /gallery — sharable, anonymously browsable.

Component endpoint: POST /api/components with {name, description, thumbnail_base64, bbox_mm...}.

Source: frontend/src/pages/Gallery.jsx renders the grid; frontend/src/components/ShareDialog.jsx is the publish flow; backend lives at backend/routes/gallery.py.

Voice triggers: 'Open share dialog' / 'Save as component'.

*Tutorial generated May 29, 2026 from ForgeSlicer source. Run the corresponding scripts/build*_tutorial.py to regenerate after future updates.*